

Supplier integration in complex delivery projects: Comparison between different buyer–supplier relationships

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Abstract

Research on project procurement management has centered on identifying the right suppliers and managing contracts to allocate risks and responsibilities. However, working with suppliers requires continuous integration activities from the buyer during project execution. This paper asks how buyers integrate their suppliers to the project organization during project execution, and how different types of buyer–supplier relationships differ in their supplier integration. We collected interview data in two complex delivery projects, both unique in their buyer–supplier relationships. We explored supplier integration practices from both the buyer's and the supplier's perspective, and sought cross-case differences possibly attributable to the different inter-organizational relationships. Based on the analysis, the nature of the buyer–supplier relationship appears to be associated with the type of integration practices used. The results of the study have implications on how supplier integration should be configured in different buyer–supplier relationships.

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1. Introduction

In project-based industries, delivering complex projects to clients requires that main contractors purchase sub-projects and expertise from external suppliers. Project procurement management focuses on planning acquisitions, identifying and choosing the right suppliers, planning and negotiating appropriate contracts, and administering and closing contracts (Cox and Thompson, 1997; Turner and Simister, 2001; Project Management Institute, 2004). Besides the aforementioned activities, working with suppliers includes various day-to-day activities to ensure that planned work is carried out as agreed. This paper is concerned with such activities that integrate the supplier to the project organization during project execution.

Supplier integration is here defined as collaboration and control between the project contractor and its supplier dur-

ing project execution. Extant literature has examined supplier integration and its mechanisms in various business contexts such as manufacturing (Swink et al., 2007), product development (Stock and Tatikonda, 2004; Petersen et al., 2005; Koufteros et al., 2007; Song and Di Benedetto, 2008), complex products and systems (Hobday, 2000; Davies, 2004; Hobday et al., 2005), and services (Edwards and Olsson, 1996). The effects and benefits of integrative mechanisms have received considerable attention. The viewpoint of the buyer has been emphasized as compared to the suppliers' viewpoint.

While extant literature on supplier integration has significantly advanced knowledge on both the means and the importance of integrative practices, it still lacks empirical research carried out in the context of complex project deliveries. Complex projects form the primary mode of organizing production in many of today's project-based industries such as construction (Eccles, 1981), shipbuilding (Tikkanen, 1997), fashion (Uzzi, 1997), oil and gas (Olsen et al., 2005), sport events and festivals (Pitsis et al.,

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2003), and media (Miles and Snow, 1986; Windeler and Sydow, 2001). In addition, extant research on supplier integration has directed only limited attention to exploring how the use of integrative practices may be influenced by the relationship between the main contractor and the supplier. Extant research focusing on inter-organizational relationships has discussed how relationships develop over an extended period of time and how commitment and trust may facilitate cooperation and provide opportunities for learning and innovation (Powell, 1990; Heide and Miner, 1992; Dyer, 1997).

The objective of this paper is to increase understanding on how project contractors integrate suppliers as a part of the project organization during project execution. In particular, we investigate complex delivery projects and the role of buyer–supplier relationships with regards to supplier integration. Our empirical case study is guided by two research questions:

1. With what types of practices do project contractors integrate their suppliers to complex delivery projects during project execution?
2. How do integration practices differ across projects that differ from each other in the duration and commitment of their buyer–supplier relationship?

2. Literature review

In several industries such as construction, engineering, shipbuilding, and television production, deliveries are generally organized as large complex projects that include a complex combination of tangible products and intangible services (Hobday, 1998; Davies, 2004; Hobday et al., 2005). The production of complex products and systems is typically organized as temporary inter-firm projects and project networks (Winch, 1989; Hellgren and Stjernberg, 1995). Project contractors are increasingly concentrating on their core activities while serving their clients, and may outsource other activities to external suppliers (Miller et al., 1995). The contractor often acts as a systems integrator and takes responsibility for actively coordinating a network of upstream suppliers and subcontractors (Davies, 2004; Hobday et al., 2005).

Literature on systems integration and project-based firms acknowledges that integration is required between the firm and its stakeholders. Project-based firms must integrate components, skills and knowledge from other organizations to produce complex solutions (Hobbs and Andersen, 2001; Hobday et al., 2005), and the success of delivering complex solutions depends on the efficient functioning of the entire inter-organizational network (Gann and Salter, 2000). How suppliers are monitored, controlled and integrated in this network is likely to influence the success of the entire project. For example, some of the features of complex systems require the contractors and their suppliers to continue negotiations on the features of the solu-

tion beyond initial contract, throughout the design and delivery of the solution (Hobday, 1998). Also, contractors and their suppliers must continually interact to integrate new knowledge coming from both the client and other stakeholders (Gann and Salter, 2000). The complexity of the delivered system, therefore, requires that suppliers are well integrated in the project not just in contract negotiations, but also during its deployment. In this paper, supplier integration is defined as collaboration and control between the project contractor and its supplier during project execution.

2.1. Supplier integration in complex projects

In complex delivery environments, project-based organizing is seen as an effective way to integrate different types of knowledge and skills across boundaries of several firms to achieve the complex product or system (Hobday, 2000; Gann and Salter, 2000; Brady et al., 2005). Some complex projects involve so many suppliers and stakeholders that the integration task requires a separate project office to deal with various integration challenges (Shenhar et al., 2001). Expertise and expert service centered supplier integration has increasingly been discussed in the context of product development where the degree of uncertainty and technical novelty is typically high, each project is unique and requires a unique configuration of internal and external capabilities.

The need to coordinate external suppliers' and customers' work has been emphasized during product development projects (Millson and Wilemon, 2002; Petersen et al., 2005; Koufteros et al., 2007). Supplier integration is needed to bring in more and better knowledge to the project as early as possible (Petersen et al., 2005; Koufteros et al., 2007; Song and Di Benedetto, 2008), to improve project performance and the quality of the solution (Primo and Amundson, 2002; Ragatz et al., 2002; Koufteros et al., 2007), to generate innovations (Koufteros et al., 2007; Song and Di Benedetto, 2008), to shorten the duration of the project (Hartley et al., 1997), and to co-invest and share risks in the venture (Song and Di Benedetto, 2008). However, the anticipated benefits of supplier integration have not always been supported (Hartley et al., 1997), and also the costs of integration in the form of required effort are frequently mentioned (Primo and Amundson, 2002).

How and through which practices should project contractors integrate suppliers into the project, that is, to collaborate with and to control them? Prior research has directed plenty of attention to supplier assessment and selection (Hartley et al., 1997; Stock and Tatikonda, 2004; Petersen et al., 2005; Ledwith and Coughlan, 2005), timing of supplier involvement (Primo and Amundson, 2002; Petersen et al., 2005; Song and Di Benedetto, 2008), and the nature, quality and embeddedness of the buyer–supplier relationship (Primo and Amundson, 2002; Stock and Tatikonda, 2004; Ledwith and Coughlan, 2005; Petersen et al., 2005; Koufteros et al., 2007; Song

and Di Benedetto, 2008). Prior research also indicates that the success of supplier integration is contingent on factors such as technical difficulty or novelty of the project (Primo and Amundson, 2002; Ragatz et al., 2002), supplier's role (Petersen et al., 2005), firm size (Ledwith and Coughlan, 2005; Koufteros et al., 2007), national context (Ledwith and Coughlan, 2005), and contractor's relative power and commitment to the supplier (Song and Di Benedetto, 2008). Many of the integration mechanisms and contingency factors associated with success deal with knowing the supplier well, aligning the roles properly, and characteristics of the buyer–supplier relationship, which may not be achievable within one single project but may require a longer term commitment both from the contractor and the supplier.

The above mentioned studies focus on fairly abstract and momentary concepts of involvement, embeddedness, interaction resources, and relationship quality that do not really tell about the mundane practices used to collaborate with and control the supplier. Prior research suggests that both control and collaboration-oriented aspects of integration are needed (Turner and Müller, 2004; Caglio and Ditillo, 2008), but does not adequately consider how the integration mechanisms differ in nature and what types of integration mechanisms are particularly suited for certain contexts. In particular, Caglio and Ditillo (2008) claim that in highly uncertain transactions, risky environments and symmetrical partnerships, formal market and hierarchy based controls are not sufficient but, additionally, informal and collaborative forms of integration – including trust and decentralized cooperation – are needed.

2.2. Buyer–supplier relationships in complex projects

Relationships play a considerable role in determining how different forms of collaboration, including all economic exchange, are carried out between involved actors (Ouchi, 1980; Jarillo, 1988; Noordewier et al., 1990; Dyer, 1997; Zaheer et al., 1998; Zaghoul and Hartman, 2003; Cannon et al., 2002). The role of relational ties in supplier integration appears as central: previous research argues that buyer–supplier relationship is associated with integration success (Primo and Amundson, 2002; Stock and Tatikonda, 2004; Ledwith and Coughlan, 2005; Petersen et al., 2005; Koufteros et al., 2007; Song and Di Benedetto, 2008).

A number of studies have examined inter-organizational relationships between actors engaged in the delivery of large and complex projects. Eloranta (2007) demonstrated how the discontinuous nature of demand in many project-based industries results in difficulties for project actors to develop relationships characterized by trust and commitment. This difficulty is imminent in relationships between clients and main contractors as it is typical for the client not to purchase another project from the same contractor for several years (Hadjikhani, 1996). There are also project-based industries such as construction where the main contractor may be able to work with its subcontractors

on a more recurrent basis, jointly delivering projects for different clients (Eccles, 1981). Söderlund and Andersson (1998) have argued that the lack of established relationships between project participants can reduce the probability of project success.

A significant amount of research on inter-organizational relationships has focused on identifying relevant dimensions of relationships and categorizing them accordingly. Dimensions often argued as central for describing the characteristics of inter-organizational relationships include: duration, commitment, trust, interdependence, goal congruence, presence of social bonds, contractual coordination mechanisms, and adaptation (Thorelli, 1986; Johanson and Mattsson, 1987; Ring and Van de Ven, 1992; Grandori and Soda, 1995; Dyer, 1997; Uzzi, 1997; Oliver and Ebers, 1998).

Commitment develops between actors when they over time come to care about the organizations or the individuals employed by them and cooperate out of altruism rather than exogenous requirements (Macneil, 1978; Heide and Miner, 1992). A high degree of commitment between the buyer and the supplier can facilitate cooperation and limit opportunistic behavior (e.g. Powell, 1990; Heide and Miner, 1992). As supplier integration can be considered as an important component of buyer–supplier cooperation, we expect that the degree of commitment of the buyer–supplier relationship may affect both the mechanisms and outcomes of supplier integration in the context of complex projects. Duration has been identified as another important dimension characterizing inter-organizational relationships (Eccles, 1981; Ring and Van de Ven, 1992; Gulati 1995; Uzzi, 1997). More specifically, as two actors collaborate over an extended period of time, the relationship between them becomes increasingly institutionalized, which tends to lead to the development of shared routines and practices that enable the involved actors to collaborate more effectively (Eccles, 1981; Uzzi, 1997). Following this reasoning, we expect that the temporal duration of the buyer–supplier relationship may affect both the mechanisms and outcomes of supplier integration in the context of complex projects.

3. Research strategy and cases

We employed a qualitative, comparative two-case strategy to respond to the research questions. We sought for projects that would represent complex system deliveries where the main contractor and its supplier are involved in a buyer–supplier relationship characterized by either a high degree of commitment or long temporal duration. In addition, we sought for cases that would be illustrative of supplier integration during project execution in technology-centered project-based industries of construction, engineering and shipbuilding. As a result, we ended up with projects “SafetySystem” and “DesignSolution”. These two cases and their inherent differences are illustrated in Table 1. For reasons of confidentiality, the cases and involved firms need to be kept anonymous.

Table 1
Description of contractors, suppliers and the case projects between them.

	SafetySystem	DesignSolution
Contractor	Company A • A large, local unit of a global firm specializing in one main business • A large, technology-based and service-supported complex system as a product • Typical project deliveries last for several years, involve hundreds of people and consist of hundreds of sub-projects	Company B • A large, local unit of a global and diversified firm • A variety of different services and products as well as complex systems containing both services and products • Typical project deliveries vary from small services and technical system deliveries to large and complex project deliveries
Supplier	Company C • A global medium-sized enterprise specializing in a specific technology-centered safety related solution	Company D • A large global and diversified firm, with system design capabilities in one of Company B's businesses
Case project	• System delivery • Represents a minor part of the entire project delivery towards the contractor's client • Highly dependent on other subsystem deliveries • Covers the entire delivery lifecycle	• Entire design solution, including system delivery • A significant part of the entire complex system delivery to the contractor's client • Other subsystems are highly dependent on the design solution • Covers the entire delivery lifecycle
Buyer–supplier relationship	• Long temporal duration • Low to moderate degree of commitment	• Short temporal duration • High degree of commitment

Project “SafetySystem” is a system delivery between the main contractor Company A and its supplier Company C. Company A has full responsibility for delivering the project for the client and it subcontracts the majority of the work and components from hundreds of suppliers. Company C delivers one subsystem that is highly complex, strongly dependent on other subsystem deliveries, and connected to all the major parts included in the delivery to the client. Both Company A and Company C have allocated their own project managers in SafetySystem project, and Company C takes full responsibility for its subcontracted scope of work. Both companies operate in the same country, partially with foreign personnel.

Project “DesignSolution” can be characterized as a partnership project between Company B and Company D as the companies have formally agreed on alliance cooperation. Company D delivers a system design as part of Company B's project with its client, and both companies operate as members in a shared project organization and collaborate directly with Company B's client. The project is managed by a project manager employed by Company B. The companies operate in two different European countries, Company B quite closely located with the client. Fig. 1 illustrates the project organizations of the studied cases.

The buyer–supplier relationship between companies A and C is characterized by a long temporal duration as the companies A and C have collaborated for over a decade, always on a project-by-project basis. Company B and Company D have collaborated on a very limited number of projects, and the buyer–supplier relationship between them is characterized by a short temporal duration. Despite the short history together, the companies have agreed on a strategic partnership which signifies mutual commitment to work together in multiple consecutive projects.

3.1. Data collection and analysis

Data were collected by 15 semi-structured interviews with key persons involved in the case projects: project managers, coordinators, lead engineers and supervisors both from the contractor's and the supplier's side. Seven of the interviews represented the SafetySystem project (four persons from Company A and three from Company C) and eight the DesignSolution project (five persons from Company B and three from Company D). The informants for the interviews were chosen by internal experts employed by companies A and B, with the intent of finding knowledgeable, experienced individuals who would have first-hand knowledge on supplier integration, both from contractor's and supplier's side.

The interview protocol covered seven topic areas including the respondent's background, the firm's own supervision practices, general issues in subcontracting relationship, and specifics of supplier integration, the project, and improvement needs. We particularly center to supplier integration in the case projects, and respondents' perceptions of the buyer–supplier relationships between the main contractor and the supplier. The interviews lasted between 27 and 110 min, averaging 59 min. A majority of the interviews were held face-to-face, with four exceptions in which the interviews were held by telephone (all Company D interviewees and the site supervisor in Company C). All interviews were recorded and transcribed to improve the reliability of analysis.

For the purpose of the analysis, we first explored the data by reading the transcripts, identifying areas in which supplier integration and buyer–supplier relationships were covered. Secondly, we developed a coding scheme based on the literature review and interview content, covering several aspects of supplier integration (dependencies, partner selection, boundary objects, meetings,

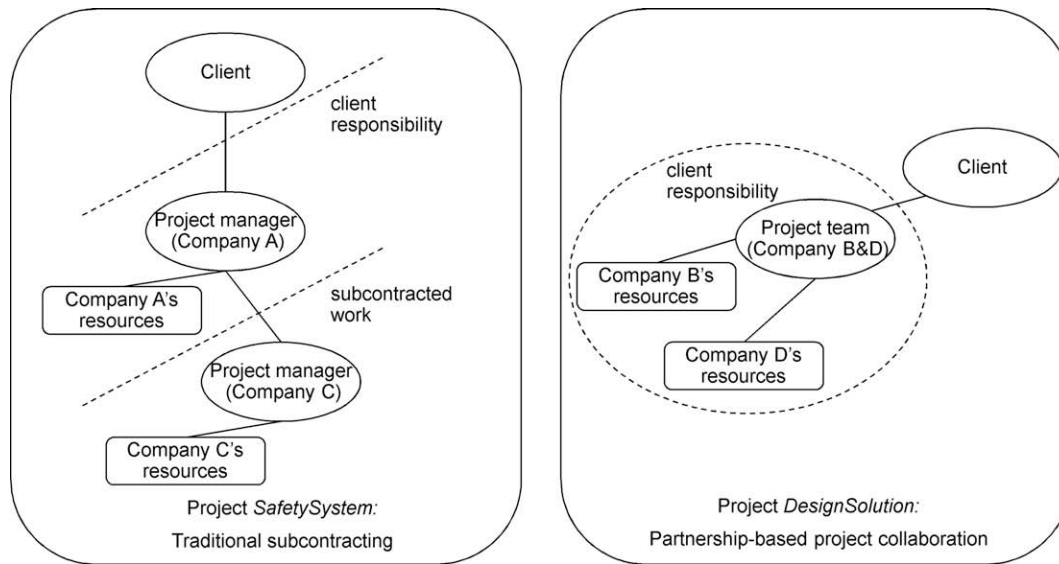


Fig. 1. Cases: project SafetySystem is a subcontracted system delivery, and project DesignSolution is a collaborative solution design project.

individual integrators, informal cooperation, reviews, and team processes) and buyer–supplier relationships (history, current state, future expectations). Thirdly, we coded and cross-tabulated the data for supplier integration and buyer–supplier relationships separately for both the contractor and its supplier in the two cases. In accordance with Miles and Huberman (1994) we also sought for repeated patterns within each case, identified discrepancies between studied actors, and looked for cross-case similarities and differences. Finally, we reflected the case findings on key issues in the literature review. We will first report case-specific results and then a cross-case analysis.

4. Results

4.1. SafetySystem project

Company C is one among hundreds of suppliers for Company A, but only a few other firms are able to offer deliveries that are similar in scope as Company C's offerings.

Company A purposefully uses competitive bidding to keep the price level low, but it has repeatedly chosen Company C due to the high reliability and delivery quality compared to competitors. Over a period of over 10 years, Company C has delivered several projects to Company A. For the project SafetySystem, Company A had requested Company C to take responsibility for a slightly broader and more complex system delivery than in prior projects. According to the interviewees, the integration challenge between companies A and C can be characterized as a technical and system integration challenge where SafetySystem is dependent on requirements and input information from Company A and its other suppliers.

4.2. Integration mechanisms in SafetySystem

Supplier integration in SafetySystem relied strongly on five mechanisms. Firstly, the project had a broker or integrative person, i.e. consultant working for Company C located in Company A premises. This role was used by Company C for the first time due to the broad scope of SafetySystem project. All interviewees felt that the use of a broker was a good practice and had helped keep the project on track in a significant way. Also some other integrative persons such as project managers and lead engineers were mentioned. Secondly, many kinds of boundary objects (i.e. objects that are shared and shareable across different problem-solving contexts, Star, in Carlile, 2002) were in place, ranging from project contract and plan to drawings, sketches, norms, installation instructions, material handling systems, and documentation systems. The computerized systems for materials, documentation, and operations were seen as crucial in the complex organization, for everyone to know the status of different tasks. Thirdly, the project relied on regular, continuous monitoring where progress was compared to plans and documents. Monitoring concerned materials, work, documents, progress, and deviations from plan, and was done by the project managers, Company A's client, and an external verification organization.

Fourthly, a variety of meetings were mentioned as an integrative mechanism. Both regular and ad hoc review meetings took place at the different levels of the organization. Some of the interviewees also mentioned that training events and information sessions had been organized because of the special features in the scope of SafetySystem, to instruct employees on the novel requirements. Finally, informal interaction formed an important part of supplier integration. Interviewees mentioned that e-mail

and phone conversations took place daily and that interaction was particularly active when problems or deviations occurred. Much of integration depended on trust between the partners and reliance on individual's capabilities to do their jobs. Despite the presence of trust between Companies A and C, both parties emphasized difficulties related to informal collaboration. "It is like we are on the edge of a cliff all the time, we must demand something from [the supplier] constantly", as the installation supervisor of Company A said. The site supervisor of Company C felt that the informal approach caused many practical problems: e.g. "Just a short while back we had a situation where we had nothing on paper and they interpreted one thing and we interpreted it totally differently".

Interviewees in both companies also mentioned practices that had been intended to support integration in SafetySystem but had not functioned satisfactorily. For example, both companies had their own material coding and documentation systems and different ways of blueprinting the system. However, some improvements had already been made for SafetySystem project, e.g. Company A had added Company C's material codes to their own coding system which helped both companies in interpreting partial deliveries of the system.

4.3. Buyer–supplier relationship in SafetySystem

SafetySystem is one project in a continuum of almost ten consequent projects, delivered by Company C to Company A over a period exceeding ten years. Over this extended period of time, an established relationship had formed between the two companies and all interviewees were aware of it. Many reported that the current practices of working were a partial consequence of learning in the prior joint projects. Despite the long temporal duration of the relationship between Companies A and C, commitment in the relationship between the two companies can be described as relatively low as interviewees representing Company A did not consider to be more committed to working with Company C than any other Company A's supplier in the future.

The interviewees in both companies reported that collaboration in general was functioning well, they had high reliance towards the partner company, the other company was already well known and "does its job rather well". Where Company A appreciated the high quality and trustworthiness of their supplier, Company C valued the scope of the business and flexibility of Company A. Both companies also expressed some worries with regards to the collaboration, especially in terms of lack and turnover of competent personnel in the other firm and problems with material deliveries. Company A interviewees mentioned that sometimes the supplier "tries to minimize its own work to make immediate profit", whereas Company C interviewees mentioned A's tendency "to rush and mess things up when the schedule becomes critical", which causes confusion in the large organization.

The fragmented organization of Company A, the length of decision chains, and turbulence among the international installation personnel were mentioned by both firms' interviewees as issues making collaboration difficult. Finally, the special nature of SafetySystem project for the development of the buyer–supplier relationship between companies A and C was clearly acknowledged. SafetySystem namely meant an increased responsibility for Company C in controlling its own work, and also the requirement to develop new procedures and practices. Both partners saw SafetySystem as an important step on the way towards Company C offering an integrated solution for Company A, significantly exceeding the scope of any of Company C previous deliveries.

4.4. DesignSolution project

Companies B and D shared only a short history of working together but had already agreed on a strategic partnership, demonstrating commitment to engage in joint projects in the future. Common ways of working between the two companies had just started to develop. The integration challenge and dependency between the firms in DesignSolution project was clearly expressed as the need to serve Company B's client by operating as one unified front and thereby fulfilling the specifications, design requirements and schedules obtained from the client.

4.5. Integration mechanisms in DesignSolution

Supplier integration in DesignSolution relied primarily on six mechanisms. Firstly, the project had an integrated engineering team that collaborated in all areas of the project, ranging from the original customer offer and contract negotiations to solution design, system delivery, and commissioning. The team had its own project manager who was employed by Company B, its own project organization, common activities and procedures, and day-to-day management. Secondly, much of integration was associated to the work of multiple project managers who lead the (sub-)projects carried out by the integrated engineering team. Also, lead engineers and the project's steering committee were mentioned as integrative parties.

Thirdly, boundary objects were seen as important instruments of integration. All interviewees talked about the alliance agreement between companies B and D. Additionally, project plan, documentation, schedules, reports, each organization's own project management system, design documents and engineering tools were in use. Fourthly, different meetings were used both in a regular and ad hoc manner. In particular, the project kickoff was mentioned as an event where the project team took shape and many important issues were agreed on.

Fifthly, irregular monitoring at the different phases of the project took place in project DesignSolution. A differentiation was made between the cost-reimbursable parts of the project and lump-sum parts, i.e., where the

lump-sum-based system delivery was monitored more as inspections of deliverables, the cost-reimbursable engineering and commissioning work was monitored as monthly reviews and analysis of deviations. Further, although the project manager was in charge of monitoring the progress of the project as a whole, each of the two companies also had complementary proprietary systems of monitoring progress. Finally, the role of informal interaction was emphasized in agreeing, controlling, and steering the work in DesignSolution. Various means of day-to-day communication, reliance on individual's competence, co-location with the client, and personal interaction were emphasized as ways of supporting integration. As both companies B and D had their own project reporting systems and had not agreed on shared practices, they had to “just rely on what Company D tells us and trust them”, as a lead engineer in Company B described.

4.6. Buyer–supplier relationship in DesignSolution

DesignSolution project is the third joint project between companies B and D and as a result, the temporal duration of the buyer–supplier relationship between the two companies is still limited. Companies B and D have, however, committed to collaborate in future projects bringing new business for both firms. According to the interviews, an important rationale for the partnership between the two firms stems from the complementarity of the competence bases and both firms' interest to generate new business in the client's industry.

The limited temporal duration of the relationship between companies B and D, and the resulting lack of experience in working together became evident during the interviews. Representatives of both companies expressed that the companies were very different from each other, both in terms of business and culture. In addition, interviewees characterized joint projects between companies D and B more as subcontracting than partnership because of Company D's unfamiliarity with the core business of Com-

pany B's clients. Culturally, the differences were seen as a challenge in day-to-day activities and communication. “We [Company D] are more a procedural company, and they [Company B] are more an entrepreneurial company”, as one of Company D's project managers put it.

DesignSolution had some implications for the development of the relationship between the two companies, as the project had changed Company D's role from a reactive system supplier to an active participant. The project clearly meant a need for the companies to work together in integrated teams and for Company D to take a more visible role towards the client. Although interviewees in both firms reported quite a high distance in locational, cultural and procedural sense between the companies, they at the same time expressed both the willingness and need to develop common guidelines and practices for the collaboration.

4.7. Cross-case analysis

To further analyze the data, we clustered the identified supplier integration practices into control-related and cooperation-related mechanisms. Table 2 summarizes key differences in integration mechanisms identified in comparing the two cases.

Although both control and cooperation-oriented integrative mechanisms were identified in both projects, our results point out clear differences between the cases. In SafetySystem, the interviewees emphasized the use of continuous and regular control-oriented integrative activities. The buyer–supplier relationship, built over a period of time exceeding 10 years, had led to the development of mutual trust that functioned as an integrative mechanism, but was clearly complemented with continuous controls, practical instructions and guidelines that had been developed based on learnings from prior projects between the companies A and C. Company A had a dominating role in the relationship and largely laid the rules by which Company C operated.

Concerning the DesignSolution project, the emphasis was on cooperation-oriented activities. The commitment

Table 2
Cross-case analysis of supplier integration.

	SafetySystem (companies A and C)	DesignSolution (companies B and D)
<i>Integration mechanisms emphasizing control</i>		
Supplier selection	– Quality, reliability, continuity and customer request as bases for supplier selection	– Competence complementarity as a basis for supplier selection
Boundary objects	– Attempts to align boundary objects, e.g. similar coding and shared systems	– Cooperation agreement – Expressed need for practical guidelines and tools to instruct day-to-day operations
Monitoring of supplier	– Formal, continuous and regular	– Irregular and project phase-specific – Differentiated monitoring of cost-reimbursable and lump-sum part of the project
<i>Integration mechanisms emphasizing cooperation</i>		
Integrative persons	– Broker (coordinator) representing Company C	– Project managers: shared project manager and firm-specific sub-projects' project managers
Team processes	– Informal interaction – Training and instruction – Regular and ad hoc meetings	– Integrated engineering team – Informal interaction – Regular and ad hoc meetings
Trust	– Primarily competence-based trust	– Primarily contract-based trust

formally expressed as a rather general alliance agreement had opened up an arena for interaction and knowledge sharing which largely relied upon working together in an integrated team, not actively using formal or structured control mechanisms. The presence of ad hoc forms of reviews indicated that working practices between the two companies had not yet become routine. Interviewees expressed their wish that some common guidelines and methods would be developed in the future. The external client had a dominating role in defining the rules for the relationship between Companies B and D.

5. Discussion

5.1. *Supplier integration in complex delivery projects*

In the first research question we inquired about practices of supplier integration in complex delivery projects. Based on the data from two complex delivery projects we identified control-oriented practices of supplier selection, using boundary objects, and monitoring, and cooperation-oriented practices of using integrative persons, team processes and trust. Our results lend general support to supplier integration literature in product development contexts that directs attention to supplier assessment and selection, team processes, relationship characteristics, and quality control (Hartley et al., 1997; Ledwith and Coughlan, 2005; Petersen et al., 2005; Koufteros et al., 2007). In addition, we have drawn attention to managing the day-to-day supplier relationship actively also during project execution, and not only during contract negotiations and administration.

In particular, we have contributed by opening up the “black box” of supplier integration in the context of complex products and systems. Earlier research on complex products and systems has emphasized the relevance of supplier integration (Hobday, 2000; Gann and Salter, 2000), but has not specified how such integration actually takes place. Compared to supplier integration in product development, our data in complex projects revealed the relevance of using integrative persons, boundary objects and cross-organizationally integrated engineering teams. Integration, therefore, was shown in a rather operative and practical light and not only as a strategic issue of capability evaluation, involvement timing, and relationship features. Based on our findings, even with contractual commitment in place, supplier integration requires the active effort of project owners, managers and personnel throughout the project execution in creating practical roles, events and tools for integration.

5.2. *Buyer–supplier relationships and supplier integration*

The second research question inquired how integration practices differed in different buyer–supplier relationships. In particular, we investigated two polar cases, a buyer–supplier relationship characterized by long temporal duration combined with a low degree of commitment, and a

buyer–supplier relationship characterized by short temporal duration combined with a high degree of commitment. Our results illustrate that in the long-lasting relationship, integration was dominated by control-oriented activities, whereas in the commitment-based relationship, integration practices were more cooperation-centered.

Based on this finding, we suggest that temporal duration and degree of commitment in a buyer–supplier relationship may be associated with the types of integrative mechanisms used in integrating the supplier to the project organization. Our finding contributes to the literature on inter-organizational relationships that explores economic exchange between organizational actors. In particular, earlier studies have shown how relationships between economic actors can enable the use of social coordination mechanisms (Jones et al., 1997), facilitate sharing of fine-grained information and joint problem-solving activities (Uzzi, 1997), and lead to increased performance and reduction of transaction costs (Ouchi, 1980; Jarillo, 1988; Noordewier et al., 1990; Cannon et al., 2002). The analysis in SafetySystem and DesignSolution projects revealed alternative configurations of control and collaboration-oriented integration mechanisms used in the different types of relationships.

The findings have implications for research on inter-organizational relationships in project-based industries. Hadjikhani (1996) and Eloranta (2007) have discussed how the discontinuous nature of demand in project-based industries causes difficulties for project actors to develop relationships characterized by trust and commitment. Based on our findings, it is likely that commitment through cooperative integration practices may be short lived if the lack of control endangers the results of the project. On the other hand, discontinuities may affect supplier integration negatively as the buyer in repeated relationships might be prone to choosing overly control-oriented mechanisms for supplier integration. Our findings thereby provide support to earlier research (e.g. Eccles, 1981; Dubois and Gadde, 2000) that has shown how both repeated exchange and time are required to develop routine working practices between the buyer and the supplier in many project-based industries. In the observed buyer–relationship characterized by a short temporal duration, many integrative mechanisms were still rather experimental or ad hoc in nature, and the monitoring of the supplier was dependent on the cost-logic of the subproject contract. This finding is in accordance with Hellgren and Stjernberg (1995) who have demonstrated how actors involved in complex projects continuously balance between their short-term and long-term objectives.

6. Conclusions

In this paper we examined how suppliers are integrated to the project organization when delivering complex systems in a buyer–supplier relationship. By empirically observing two projects from the viewpoint of both the buyer and the supplier, we identified several mechanisms

to achieve supplier integration and categorized these mechanisms as practices emphasizing control and those emphasizing cooperation.

By focusing on the characteristics of the buyer–supplier relationships we identified considerable differences in the use of integrative mechanisms in the two-case projects. It appears that commitment in the buyer–supplier relationship may be positively related to the use of cooperation-oriented integration activities. Temporal duration of the relationship, in turn, appeared more inclined towards increasingly routine, control-oriented integrative mechanisms.

Our findings have some implications for practitioners involved in the production of complex systems. First, based on our results, the relationship between the buyer and the supplier should be taken into consideration when selecting integration mechanisms. Second, as commitment and trust have been shown to require time and repeated exchanges to develop, the use of integrative mechanisms between a buyer and a supplier can be expected to evolve across consequent projects. Finally, practitioners involved with the delivery of complex products and systems may benefit from understanding the differences between cooperation and control-oriented integration mechanisms, and from using both types of mechanisms complementarily.

As the empirical analysis focused on two cases involving a total of four organizations, the results are not directly generalizable to other contexts. This study is limited in that, due to the methodology applied, no claims concerning causalities between supplier integration, buyer–supplier relationships, and contextual issues can be made. Therefore, we welcome additional research that would examine to what extent integrative mechanisms may be determined by the nature of the buyer–seller relationships and to what extent the development of the relationship is influenced by the integrative mechanisms in place. Furthermore, our data were restricted in their cultural and industrial contexts, and the choice of informants and the interview-centered data collection method may have generated both subjective and positive bias in the collected data. We are aware of these limitations which can be considered somewhat typical to qualitative research, and suggest further research both to overcome these limitations and to broaden understanding on supplier integration in complex projects.

Further research is required, additionally, to understand how integrative mechanisms differ across various buyer–supplier contexts, and how they are used in a broader network of suppliers. Extending the scope from the concept of supplier integration to the more formal aspects of contracting could open up avenues for further research as well, and thereby help in solving potential interest conflicts and hazards between the buyer and supplier. Also, some of our results indicated that the role of the (external) client and the client relationship should be studied further in connection with buyer–supplier relationships. Furthermore, research is needed to explore the role of both technical and market uncertainties as well as other contingencies in the supplier integration of complex delivery projects.

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